Small Dehn surgeries and SU(2)

Fan Ye

Peking University

ICM 2026

Background: The Kirby Problem

Kirby Problem 3.105 (A)

- When is a 3-manifold Y $SU(2)$ non-abelian?
- Is any integer homology sphere ($\mathbb{Z}HS^3$) $SU(2)$ non-abelian?

Definition

A manifold Y is $SU(2)$ **non-abelian** if there exists a homomorphism

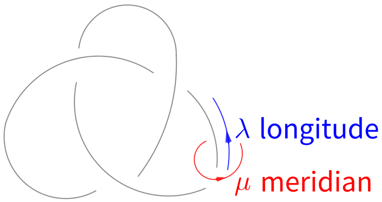
$$\rho : \pi_1(Y) \to SU(2)$$

such that $\mathrm{Im}(\rho)$ is non-abelian. Otherwise, it is called $SU(2)$ **abelian**.

Previous Results

Theorem (Kronheimer–Mrowka '04)

For any nontrivial knot $K \subset S^3$ and rational $r \in [0, 2]$, the (Dehn) surgery manifold $S^3_r(K)$ is $SU(2)$ non-abelian.

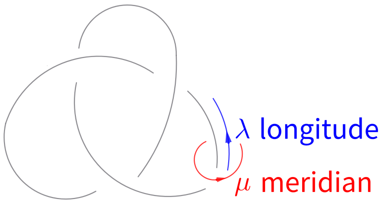


Right-handed trefoil $K = T_{2,3}$

Previous Results

Theorem (Zentner '18)

The splicing manifold $Y(K_1, K_2) = (S^3 \backslash \nu(K_1)) \cup_\partial (S^3 \backslash \nu(K_2))$ (gluing meridian to longitude and vice versa) is $SU(2)$ non-abelian.



Right-handed trefoil $K = T_{2,3}$

Remarks

Theorem (Kronheimer–Mrowka '04)

For any nontrivial knot $K \subset S^3$ and rational $r \in [0,2]$, the (Dehn) surgery manifold $S^3_r(K)$ is $SU(2)$ non-abelian.

Theorem (Zentner '18)

The splicing manifold $Y(K_1,K_2) = (S^3 \backslash \nu(K_1)) \cup_\partial (S^3 \backslash \nu(K_2))$ (gluing meridian to longitude and vice versa) is $SU(2)$ non-abelian.

- Both $S^3_{1/n}(K)$ and $Y(K_1,K_2)$ are $\mathbb{Z}HS^3$.

Remarks

Theorem (Kronheimer–Mrowka '04)

For any nontrivial knot $K \subset S^3$ and rational $r \in [0,2]$, the (Dehn) surgery manifold $S^3_r(K)$ is $SU(2)$ non-abelian.

Theorem (Zentner '18)

The splicing manifold $Y(K_1, K_2) = (S^3 \backslash \nu(K_1)) \cup_\partial (S^3 \backslash \nu(K_2))$ (gluing meridian to longitude and vice versa) is $SU(2)$ non-abelian.

- Both $S^3_{1/n}(K)$ and $Y(K_1, K_2)$ are $\mathbb{Z}HS^3$.
- We only consider $r \geqslant 0$ because for the mirror knot $m(K)$ of K,

$$\pi_1(S^3_{-r}(K)) \cong \pi_1(S^3_r(m(K)))$$

Remarks

Theorem (Kronheimer–Mrowka '04)

For any nontrivial knot $K \subset S^3$ and rational $r \in [0,2]$, the (Dehn) surgery manifold $S^3_r(K)$ is $SU(2)$ non-abelian.

Theorem (Zentner '18)

The splicing manifold $Y(K_1, K_2) = (S^3 \backslash \nu(K_1)) \cup_\partial (S^3 \backslash \nu(K_2))$ (gluing meridian to longitude and vice versa) is $SU(2)$ non-abelian.

- Kronheimer–Mrowka's theorem implies $\pi_1(S^3_r(K))$ is non-abelian for $r \in [0,2]$. The case $r = 1$ confirms the **"Property P" conjecture**.

Remarks

Theorem (Kronheimer–Mrowka '04)

For any nontrivial knot $K \subset S^3$ and rational $r \in [0,2]$, the (Dehn) surgery manifold $S^3_r(K)$ is $SU(2)$ non-abelian.

Theorem (Zentner '18)

The splicing manifold $Y(K_1, K_2) = (S^3 \backslash \nu(K_1)) \cup_\partial (S^3 \backslash \nu(K_2))$ (gluing meridian to longitude and vice versa) is $SU(2)$ non-abelian.

- Kronheimer–Mrowka's theorem implies $\pi_1(S^3_r(K))$ is non-abelian for $r \in [0,2]$. The case $r = 1$ confirms the **"Property P" conjecture**.
- Zentner further showed (in the same paper) that any $\mathbb{Z}HS^3$ Y is $SL(2,\mathbb{C})$ non-abelian (based on Geometrization Theorem).

Question on Larger Slopes

Question (Kronheimer–Mrowka '04)

Is $S^3_r(K)$ always $SU(2)$ non-abelian for $r = 3, 4$?

Question on Larger Slopes

Question (Kronheimer–Mrowka '04)

Is $S^3_r(K)$ always $SU(2)$ non-abelian for $r = 3, 4$?

Remark

This is **NOT** always true for $r = 5$. For the right-handed trefoil $T_{2,3}$:

$$S^3_5(T_{2,3}) = L(5,4)$$

Here $\pi_1 = \mathbb{Z}/5$, which is $SU(2)$ abelian.

Detailed Progress

Assuming p is **prime power**. A rational slope r for $S^3_r(K)$ is $SU(2)$ non-abelian in the following cases:

- $r = 4$ and $p/q \in (2, 3)$ (Baldwin–Sivek '23)

Detailed Progress

Assuming p is **prime power**. A rational slope r for $S^3_r(K)$ is $SU(2)$ non-abelian in the following cases:

- $r = 4$ and $p/q \in (2, 3)$ (Baldwin–Sivek '23)
- $r = 3$ (Baldwin–Li–Sivek–Y. '24)

Detailed Progress

Assuming p is **prime power**. A rational slope r for $S^3_r(K)$ is $SU(2)$ non-abelian in the following cases:

- $r = 4$ and $p/q \in (2,3)$ (Baldwin–Sivek '23)
- $r = 3$ (Baldwin–Li–Sivek–Y. '24)
- $r = p/q \in (2,5)$ (Farber–Reinoso–Wang '24)

Detailed Progress

Assuming p is **prime power**. A rational slope r for $S^3_r(K)$ is $SU(2)$ non-abelian in the following cases:

- $r = 4$ and $p/q \in (2, 3)$ (Baldwin–Sivek '23)
- $r = 3$ (Baldwin–Li–Sivek–Y. '24)
- $r = p/q \in (2, 5)$ (Farber–Reinoso–Wang '24)
- $r = 5$, when $K \neq T_{2,3}$ (Li–Y. '25a)

Detailed Progress

Assuming p is **prime power**. A rational slope r for $S_r^3(K)$ is $SU(2)$ non-abelian in the following cases:

- $r = 4$ and $p/q \in (2,3)$ (Baldwin–Sivek '23)
- $r = 3$ (Baldwin–Li–Sivek–Y. '24)
- $r = p/q \in (2,5)$ (Farber–Reinoso–Wang '24)
- $r = 5$, when $K \neq T_{2,3}$ (Li–Y. '25a)
- $r = p/q,\ 2p/q \in (2,6)$, when $(K,r) \neq (T_{2,3}, 6 - 1/n)$ for $n \in \mathbb{Z}_+$ (Li–Y. '25b)

Detailed Progress

Assuming p is **prime power**. A rational slope r for $S^3_r(K)$ is $SU(2)$ non-abelian in the following cases:

- $r = 4$ and $p/q \in (2,3)$ (Baldwin–Sivek '23)
- $r = 3$ (Baldwin–Li–Sivek–Y. '24)
- $r = p/q \in (2,5)$ (Farber–Reinoso–Wang '24)
- $r = 5$, when $K \neq T_{2,3}$ (Li–Y. '25a)
- $r = p/q,\ 2p/q \in (2,6)$, when $(K,r) \neq (T_{2,3}, 6-1/n)$ for $n \in \mathbb{Z}_+$ (Li–Y. '25b)
- **Depending on preprints of Daemi–Liu–Miller-Eismeier–Scaduto not yet available:**
  $r = p/q,\ 2p/q \in (2,8]$ when $(K,r) \neq (T_{2,3}, 6-1/n)$ for $n \in \mathbb{Z}$ (Ghosh–Miller-Eismeier '25)

Detailed Progress

Assuming p is **prime power**. A rational slope r for $S^3_r(K)$ is $SU(2)$ non-abelian in the following cases:

- $r = 4$ and $p/q \in (2,3)$ (Baldwin–Sivek '23)
- $r = 3$ (Baldwin–Li–Sivek–Y. '24)
- $r = p/q \in (2,5)$ (Farber–Reinoso–Wang '24)
- $r = 5$, when $K \neq T_{2,3}$ (Li–Y. '25a)
- $r = p/q,\ 2p/q \in (2,6)$, when $(K,r) \neq (T_{2,3}, 6 - 1/n)$ for $n \in \mathbb{Z}_+$ (Li–Y. '25b)
- **Depending on preprints of Daemi–Liu–Miller-Eismeier–Scaduto not yet available:** $r = p/q,\ 2p/q \in (2,8]$ when $(K,r) \neq (T_{2,3}, 6 - 1/n)$ for $n \in \mathbb{Z}$ (Ghosh–Miller-Eismeier '25)

Tool: Framed Instanton Homology

The condition p **prime power** allows us to use framed instanton homology $I^\sharp(Y)$ (Kronheimer–Mrowka '11) defined for **ANY** closed 3-manifold Y.

Tool: Framed Instanton Homology

The condition p **prime power** allows us to use framed instanton homology $I^\sharp(Y)$ (Kronheimer–Mrowka '11) defined for **ANY** closed 3-manifold Y.

- $I^\sharp(Y)$ is a $\mathbb{Z}/4$-graded $\mathbb{Z}$-module.

Tool: Framed Instanton Homology

The condition p **prime power** allows us to use framed instanton homology $I^\sharp(Y)$ (Kronheimer–Mrowka '11) defined for **ANY** closed 3-manifold Y.

- $I^\sharp(Y)$ is a $\mathbb{Z}/4$-graded $\mathbb{Z}$-module.
- Euler characteristic (Scaduto '15):

$$\chi(I^\sharp(Y)) = \begin{cases} |H_1(Y;\mathbb{Z})| & b_1(Y) = 0 \\ 0 & b_1(Y) > 0 \end{cases}$$

Tool: Framed Instanton Homology

The condition p **prime power** allows us to use framed instanton homology $I^\sharp(Y)$ (Kronheimer–Mrowka '11) defined for **ANY** closed 3-manifold Y.

- $I^\sharp(Y)$ is a $\mathbb{Z}/4$-graded $\mathbb{Z}$-module.
- Euler characteristic (Scaduto '15):

$$\chi(I^\sharp(Y)) = \begin{cases} |H_1(Y;\mathbb{Z})| & b_1(Y) = 0 \\ 0 & b_1(Y) > 0 \end{cases}$$

Remark

Floer's original homology $I(Y)$ was only defined for $\mathbb{Z}HS^3$ with $\chi(I(Y)) = 2\text{Casson}(Y)$.

- Roughly, $I(Y)$ is related to $H_*(X(Y))$,
- while $I^\sharp(Y)$ is related to $H_*(R(Y))$,

where $R(Y) = \mathrm{Hom}(\pi_1(Y), SU(2))$ and $X(Y) = R(Y)/SU(2)$.

Proof Strategy: Spectral Sequence

Theorem (Baldwin–Sivek '23 for p/q, Li–Y. '25 for $2p/q$)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then there exists a spectral sequence:

$$H_*(R(Y)) \implies I^\sharp(Y)$$

Recall that $R(Y) = \mathrm{Hom}(\pi_1(Y), SU(2))$.

Proof Strategy: Spectral Sequence

Theorem (Baldwin–Sivek '23 for p/q, Li–Y. '25 for $2p/q$)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then there exists a spectral sequence:

$$H_*(R(Y)) \implies I^\sharp(Y)$$

Recall that $R(Y) = \mathrm{Hom}(\pi_1(Y), SU(2))$.

Corollary

In such a case (only abelian representations $\rho : \pi_1(Y) \to SU(2)$),

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

We will sketch the proof of the corollary.

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

- For $r = p/q$, $H_1(Y;\mathbb{Z}) = \mathbb{Z}/p$.

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

- For $r = p/q$, $H_1(Y;\mathbb{Z}) = \mathbb{Z}/p$.
- If all $\rho : \pi_1(Y) \to SU(2)$ have abelian images, then

$$\begin{aligned} X(Y) &= R(Y)/SU(2) \\ &= \mathrm{Hom}(\mathbb{Z}/p, SU(2))/SU(2) \\ &= \{[\rho_k]\}_{k=0,\dots,\lfloor p/2\rfloor}, \end{aligned}$$

where ρ_k is the abelian representation sending $1 \in \mathbb{Z}/p$ to

$$\begin{bmatrix} e^{2\pi ik/p} & 0 \\ 0 & e^{-2\pi ik/p} \end{bmatrix}$$

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then
$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

- The stabilizer of ρ_k is:

$$\mathrm{Stab}(\rho_k) = \begin{cases} SU(2)/\pm\mathrm{Id} & k = 0, p/2 \text{ (if } p \text{ even)} \\ U(1) & \text{otherwise} \end{cases}$$

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

- The stabilizer of ρ_k is:

$$\mathrm{Stab}(\rho_k) = \begin{cases} SU(2)/\pm\mathrm{Id} & k = 0, p/2 \text{ (if } p \text{ even)} \\ U(1) & \text{otherwise} \end{cases}$$

- The pre-images of $[\rho_k]$ in $R(Y)$ are $(SU(2)/\pm\mathrm{Id})/\mathrm{Stab}(\rho_k)$. Thus:

$$R(Y) = \mathrm{Hom}(\mathbb{Z}/p, SU(2)) = \begin{cases} 1 \text{ pt} \cup \frac{p-1}{2} S^2 & p \text{ odd} \\ 2 \text{ pts} \cup \frac{p-2}{2} S^2 & p \text{ even} \end{cases}$$

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then
$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

- The stabilizer of ρ_k is:

$$\text{Stab}(\rho_k) = \begin{cases} SU(2)/\pm \text{Id} & k = 0, p/2 \text{ (if } p \text{ even)} \\ U(1) & \text{otherwise} \end{cases}$$

- The pre-images of $[\rho_k]$ in $R(Y)$ are $(SU(2)/\pm \text{Id})/\text{Stab}(\rho_k)$. Thus:

$$R(Y) = \mathrm{Hom}(\mathbb{Z}/p, SU(2)) = \begin{cases} 1 \text{ pt} \cup \frac{p-1}{2} S^2 & p \text{ odd} \\ 2 \text{ pts} \cup \frac{p-2}{2} S^2 & p \text{ even} \end{cases}$$

- We compute $H_*(R(Y)) \cong \mathbb{Z}^p$.

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

- We compute $H_*(R(Y)) \cong \mathbb{Z}^p$.

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

- We compute $H_*(R(Y)) \cong \mathbb{Z}^p$.
- Since $\chi(I^\sharp(Y)) = |H_1(Y;\mathbb{Z}))| = p$ and the rank matches, the spectral sequence $H_*(R(Y)) \implies I^\sharp(Y)$ must **collapse**.

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y;\mathbb{Z})|}$$

- We compute $H_*(R(Y)) \cong \mathbb{Z}^p$.
- Since $\chi(I^\sharp(Y)) = |H_1(Y;\mathbb{Z}))| = p$ and the rank matches, the spectral sequence $H_*(R(Y)) \implies I^\sharp(Y)$ must **collapse**.
- Since $H_*(R(Y))$ is free, the S.S. over other coefficients also collapses. Thus, we obtain $I^\sharp(Y) \cong \mathbb{Z}^p$.

Instanton L-space

Definition

A closed 3-manifold Y with $b_1(Y) = 0$ is called an **instanton L-space** if:

$$\operatorname{rank} I^\sharp(Y) = |H_1(Y;\mathbb{Z})|$$

Instanton L-space

Definition

A closed 3-manifold Y with $b_1(Y) = 0$ is called an **instanton L-space** if:

$$\operatorname{rank} I^\sharp(Y) = |H_1(Y;\mathbb{Z})|$$

Definition

A knot K is called an **instanton L-space knot** if for some $r > 0$, $S^3_r(K)$ is an instanton L-space.

Definition

A closed 3-manifold Y with $b_1(Y) = 0$ is called an **instanton L-space** if:

$$\operatorname{rank} I^\sharp(Y) = |H_1(Y;\mathbb{Z})|$$

Definition

A knot K is called an **instanton L-space knot** if for some $r > 0$, $S^3_r(K)$ is an instanton L-space.

Remark

The notion "L-space" comes from Heegaard Floer theory (Ozsváth–Szabó) as a generalization of lens spaces.

Instanton L-space and Slope Constraints

Corollary

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then:

- Y is an instanton L-space.
- K is an instanton L-space knot.

Instanton L-space and Slope Constraints

Corollary

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then:

- Y is an instanton L-space.
- K is an instanton L-space knot.

Theorem (Baldwin–Sivek '24)

If K is an instanton L-space knot, then $S^3_r(K)$ is an instanton L-space if and only if:

$$r \geqslant 2g(K) - 1,$$

where $g(K)$ is the Seifert genus of K.

Proof Idea for Main Theorem

Goal: Prove $SU(2)$ non-abelian for $r \in (2,5)$ (consider $r \in [5,6)$ later).

Proof Idea for Main Theorem

Goal: Prove $SU(2)$ non-abelian for $r \in (2,5)$ (consider $r \in [5,6)$ later).
Contradiction: If $SU(2)$ abelian $\implies r \geqslant 2g(K) - 1$.

Proof Idea for Main Theorem

Goal: Prove $SU(2)$ non-abelian for $r \in (2, 5)$ (consider $r \in [5, 6)$ later).
Contradiction: If $SU(2)$ abelian $\implies r \geqslant 2g(K) - 1$.

- If $g(K) \geqslant 3$, then $2g - 1 \geqslant 5$. So for $r < 5$, we have a contradiction immediately.
- We only need to check instanton L-space knots of small genus.

Proof Idea for Main Theorem

Goal: Prove $SU(2)$ non-abelian for $r \in (2,5)$ (consider $r \in [5,6)$ later).
Contradiction: If $SU(2)$ abelian $\implies r \geqslant 2g(K) - 1$.

- If $g(K) \geqslant 3$, then $2g - 1 \geqslant 5$. So for $r < 5$, we have a contradiction immediately.
- We only need to check instanton L-space knots of small genus.

Small Genus Cases:

- $g = 1$: Only $T_{2,3}$ (Baldwin–Sivek).
- $g = 2$: Only $T_{2,5}$ (Baldwin–Li–Sivek–Y., Farber–Reinoso–Wang).

Proof Idea for Main Theorem

Goal: Prove $SU(2)$ non-abelian for $r \in (2, 5)$ (consider $r \in [5, 6)$ later).
Contradiction: If $SU(2)$ abelian $\implies r \geqslant 2g(K) - 1$.

- If $g(K) \geqslant 3$, then $2g - 1 \geqslant 5$. So for $r < 5$, we have a contradiction immediately.
- We only need to check instanton L-space knots of small genus.

Small Genus Cases:

- $g = 1$: Only $T_{2,3}$ (Baldwin–Sivek).
- $g = 2$: Only $T_{2,5}$ (Baldwin–Li–Sivek–Y., Farber–Reinoso–Wang).

Exceptions: These specific cases (for $T_{2,3}$ and $T_{2,5}$) are checked to be $SU(2)$ non-abelian (Sivek–Zentner '22).

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

- **Idea for slopes beyond 5:** Consider $I^\sharp(Y;\mathbb{Z}/2)$. Note that

$$\operatorname{rank} I^\sharp(Y) = \dim I^\sharp(Y;\mathbb{C}).$$

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

- **Idea for slopes beyond 5:** Consider $I^\sharp(Y;\mathbb{Z}/2)$. Note that

$$\operatorname{rank} I^\sharp(Y) = \dim I^\sharp(Y;\mathbb{C}).$$

- **Conceptual reason:** $R(Y) = \mathrm{Hom}(\pi_1(Y), SU(2))$ can contain copies of $SU(2)/\pm\mathrm{Id} = SO(3)$. 2-torsion appears in

$$H_*(SO(3);\mathbb{Z}) \cong \mathbb{Z}\oplus\mathbb{Z}\oplus\mathbb{Z}/2.$$

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

- **Idea for slopes beyond 5:** Consider $I^\sharp(Y;\mathbb{Z}/2)$. Note that

$$\operatorname{rank} I^\sharp(Y) = \dim I^\sharp(Y;\mathbb{C}).$$

- **Conceptual reason:** $R(Y) = \operatorname{Hom}(\pi_1(Y), SU(2))$ can contain copies of $SU(2)/\pm\mathrm{Id} = SO(3)$. 2-torsion appears in

$$H_*(SO(3);\mathbb{Z}) \cong \mathbb{Z}\oplus\mathbb{Z}\oplus\mathbb{Z}/2.$$

Corollary (similar proof as before)

Let p be a prime power, $r = p/q,\ 2p/q > 0$. If $Y = S^3_r(K)$ is $SU(2)$ abelian, then:

- Y is an instanton L-space over $\mathbb{Z}/2$ ($\dim I^\sharp(Y;\mathbb{Z}/2) = p$).
- K is an instanton L-space knot over $\mathbb{Z}/2$ (similar definition).

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

Theorem (Li–Y. '25)

If K is an instanton L-space knot over $\mathbb{Z}/2$, then $S^3_r(K)$ is an instanton L-space over $\mathbb{Z}/2$ if and only if

$$r > \nu_2^\sharp(K),$$

with uncertainty when $r = \nu_2^\sharp(K)$, where $\nu_2^\sharp(K) = \nu_{\mathbb{Z}/2}^\sharp(K)$ is an integer concordance invariant for any $K \subset S^3$ and $\nu_2^\sharp(K) \geqslant 2g(K)$ when K is an instanton L-space knot over $\mathbb{Z}/2$.

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

Theorem (Li–Y. '25)

If K is an instanton L-space knot over $\mathbb{Z}/2$, then $S^3_r(K)$ is an instanton L-space over $\mathbb{Z}/2$ if and only if

$$r > \nu_2^\sharp(K),$$

with uncertainty when $r = \nu_2^\sharp(K)$, where $\nu_2^\sharp(K) = \nu_{\mathbb{Z}/2}^\sharp(K)$ is an integer concordance invariant for any $K \subset S^3$ and $\nu_2^\sharp(K) \geqslant 2g(K)$ when K is an instanton L-space knot over $\mathbb{Z}/2$.

- The proof for $r \in [5, 6)$ relies on this sharper bound.

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

Theorem (Li–Y. '25)

If K is an instanton L-space knot over $\mathbb{Z}/2$, then $S^3_r(K)$ is an instanton L-space over $\mathbb{Z}/2$ if and only if

$$r > \nu_2^\sharp(K),$$

with uncertainty when $r = \nu_2^\sharp(K)$, where $\nu_2^\sharp(K) = \nu_{\mathbb{Z}/2}^\sharp(K)$ is an integer concordance invariant for any $K \subset S^3$ and $\nu_2^\sharp(K) \geqslant 2g(K)$ when K is an instanton L-space knot over $\mathbb{Z}/2$.

- The proof for $r \in [5, 6)$ relies on this sharper bound.
- $\nu_{\mathbb{Z}/2}^\sharp(K)$ has generalization over any field $\mathbb{F}$ (Li–Y. '25), with the property for mirror knot $\nu_{\mathbb{F}}^\sharp(m(K)) = -\nu_{\mathbb{F}}^\sharp(K)$.

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

Theorem (Li–Y. '25)

If K is an instanton L-space knot over $\mathbb{Z}/2$, then $S^3_r(K)$ is an instanton L-space over $\mathbb{Z}/2$ if and only if

$$r > \nu_2^\sharp(K),$$

with uncertainty when $r = \nu_2^\sharp(K)$, where $\nu_2^\sharp(K) = \nu_{\mathbb{Z}/2}^\sharp(K)$ is an integer concordance invariant for any $K \subset S^3$ and $\nu_2^\sharp(K) \geqslant 2g(K)$ when K is an instanton L-space knot over $\mathbb{Z}/2$.

- The proof for $r \in [5, 6)$ relies on this sharper bound.
- $\nu_{\mathbb{Z}/2}^\sharp(K)$ has generalization over any field $\mathbb{F}$ (Li–Y. '25), with the property for mirror knot $\nu_{\mathbb{F}}^\sharp(m(K)) = -\nu_{\mathbb{F}}^\sharp(K)$.
- $\nu_{\mathbb{C}}^\sharp(K)$ is previously studied by Baldwin–Sivek '21 and motivates $\nu_{\mathbb{F}}^\sharp(K)$. $\nu_{\mathbb{C}}^\sharp(K) = 2g(K) - 1$ for instanton L-space knot K.

Thanks for your attention.
Slides on floer.wiki/en